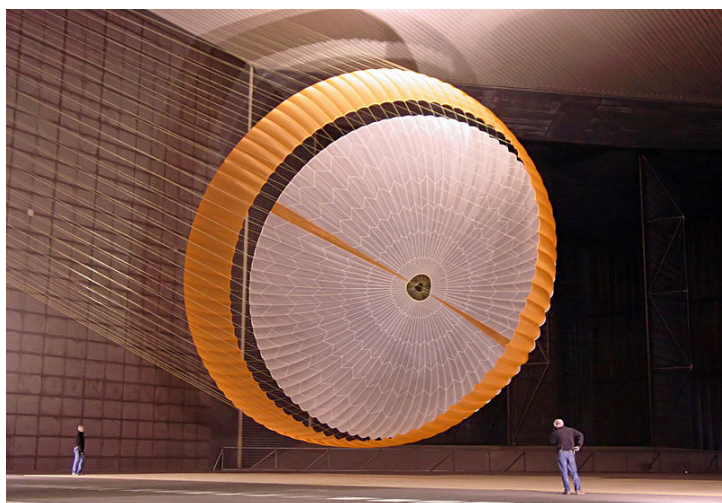
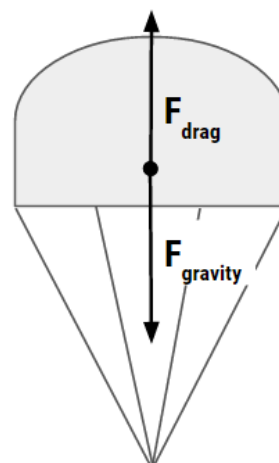


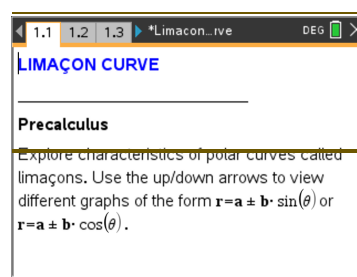
Stage 1 Physics - Parachute Investigation Design



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(Courtesy NASA/JPL-Caltech) In this activity, you will investigate the effect of changing the values of a and b in the polar equations $r = a \pm b \cdot \sin(\theta)$ and $r = a \pm b \cdot \cos(\theta)$, where $a > 0$ and $b > 0$. You will also explore the relationship between the polar curve $r = a \pm b \cdot \sin(\theta)$ (or $r = a \pm b \cdot \cos(\theta)$) and the sinusoidal function $f(x) = a + b \cdot \sin(x)$ (or $f(x) = a \pm b \cdot \cos(x)$).



Polar curves called limaçons have equations of the form $r = a \pm b \cdot \sin(\theta)$ or $r = a \pm b \cdot \cos(\theta)$. On page 1.2 are special kinds of limaçon graphs called cardioids. Use the up/down arrows to see all the graphs.

design process of the Mars Rover's parachute. Type: bit.do/marsparachute into the address bar of your browser.



Background

Spacecraft designed for re-entry on a planet that has an atmosphere such as Mars can deploy a parachute to slow the craft's rate of descent. The spacecraft falls at constant speed when the magnitude of the upward acting drag force (caused by "air" resistance) equals the magnitude of the downward acting force of gravity. The parachute is designed to minimise the speed on impact.

To minimise the force on impact you need to minimise the speed before impact. When the parachute is falling at constant (terminal) speed:

$$v_{\text{impact}} = \frac{s}{t}, \text{ so } v_{\text{impact}} \propto \frac{1}{\text{drop time}} \quad \{\text{with "s" constant}\}$$

As the relationship between v_{impact} and drop time is inversely proportional, you can minimise v_{impact} by maximising the drop time.

Video

Watch the video of the

Useful Terms

Payload: The carrying capacity of a parachute. Payload is measured in kg.

Drop time: The total time between parachute deployment and touchdown.

Suspension lines: These lines attach the parachute to the payload.

Canopy :The expanding, umbrella-like part of a parachute.

Vent :A hole in the apex to vent some air out of the canopy to reduce the oscillation.

BEFORE CONDUCTING THE EXPERIMENT

~~IAE1~~ Move to page 1.2.

~~1. Why do you think these graphs are called cardioids?~~

~~2. What similarities do you notice about the equations of the six graphs?~~

~~3. How do the addition and subtraction signs affect the graphs?~~

~~Move to page 1.3.~~

~~Use the up/down arrows to see all the graphs.~~

~~4. How are the equations different from those on page 1.2? How does this difference affect the graph?~~

~~Limaçons have different shapes depending on the ratio $\frac{a}{b}$. We have already seen the cardioid graph that is the result when $a = b$ (or $\frac{a}{b} = 1$).~~

~~Move to page 1.5.~~

RISK ASSESSMENT

Before conducting a test drop, you will write a comprehensive risk assessment and hand it in to your teacher. Graph the polar functions shown in the table by changing a and b . Complete the table with the values of a , b , and $\frac{a}{b}$ as you observe each graph. Use the up/down arrows to change the values of a and b .

INVESTIGATION DESIGN

After the test drop, you will deconstruct the question: What will affect drop time? Then design a detailed investigation to test the effect of one factor (variable) on drop time. Keep a record of your thinking during the deconstruction and annotate your design to justify the decisions you make about various aspects of the design, e.g. how and why to control variable,

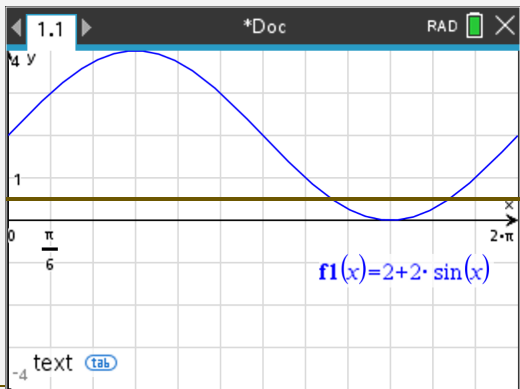
number of samples etc. Hand the deconstruction and design in to your teacher on _____.

Limaçon	a	b	$\frac{a}{b}$
i) $r1 = 1 + 3 * \cos(\theta)$			
ii) $r1 = 1 + 2 * \cos(\theta)$			
iii) $r1 = 3 + 2 * \cos(\theta)$			
iv) $r1 = 2 + 1 * \cos(\theta)$			
v) $r1 = 3 + 1 * \cos(\theta)$			

5. ~~Some of the limaçons in the table above have the ratio $\frac{a}{b} < 1$. Write an equation of another polar curve for which $\frac{a}{b} < 1$. Graph your limaçon and describe the shape of the limaçon.~~

6. ~~One of the polar curves in the table above has a ratio which satisfies $1 < \frac{a}{b} < 2$. Write an equation of another polar curve for which $1 < \frac{a}{b} < 2$. Graph your limaçon and describe the shape of the limaçon.~~

7. ~~Some of the limaçons in the table above have the ratio $\frac{a}{b} > 2$. Write an equation of another polar curve for which $\frac{a}{b} > 2$. Graph your limaçon and describe the shape of the limaçon.~~

<u>THE REPORT DUE AFTER THE EXPERIMENT</u> <u>IAE2 - IAE3 - IAE4 - KA1 - KA4</u>		Done?
<u>INTRODUCTION</u>	Physics background, hypothesis, variables	☐
	Step-by-step of the method you actually used	☐
	Present the data collected in a single Data Table of results.	☐
	Students will represent their data using a Line Graph for continuous data, or a Bar Graph for discontinuous data.	☐
	Describe the trends and relationships as shown by your data.	☐
<u>RESULTS</u>	Interpret your results with reference to science concepts. Move to page 2.2.	☐
	8.—Evaluate your data by discussing the precision (scatter) of your data, and by commenting on reliability. The graph of the sinusoidal function $f(x) = 2 + 2 \cdot \sin(x)$ on the interval $0 \leq x \leq 2\pi$ is shown below. The x scale is $\pi/6$. 	☐
	Graph the limaçon given by $r1(\theta) = 2 + 2 \cdot \sin(\theta)$. Press b and select 3 Graph Entry/Edit and then 5 Polar. Now press b and select 5 Trace and then 1 Graph Trace. Use the right arrow to move your cursor. Observe the change in the r and θ values. On the interval from $x = 0$ to $x = 2\pi$ of the sinusoidal function, the maximum occurs at $x = \frac{\pi}{2}$ ($\approx 1.6 \text{ rad}$) and the minimum occurs at $x = \frac{3\pi}{2}$ ($\approx 4.7 \text{ rad}$). How do the y-values at these two points correspond to the r-values on the cardioid?	☐
	Describe the sources of possible systematic and random exp. errors in order of greatest impact.	☐

	Evaluate your procedure.	☐
<u>CONCLUSION</u>	Write a statement in relation to your original aim and hypothesis (supports, refutes or inconclusive). This includes justification.	☐
	Describe any relationship between the independent and dependent variables established.	☐

The report should be a maximum of 1000 words if written, or a maximum of 10 minutes for an oral presentation, or the equivalent in multimodal form.

Evidence outlining the deconstruction and design process should be attached to the report. This evidence should outline the deconstruction process, the method chosen as most appropriate, and a justification of the plan of action, to a maximum of 4 sides of an A4 page. Suggested formats for the summary sheet include flow charts, concept maps, tables, or notes.

Only the following sections of the report are included in the word count:

- Introduction
- evaluation of method/procedure
- analysis of results
- conclusion.

Stage 1 Physics Rocket Science: Parachute Investigation



				A	B	C	D	E	
The deconstruction and investigation design done before the experiment including a risk assessment prepared prior to the practical.		IAE1		Critically deconstructs a problem and designs a logical, coherent, and detailed physics investigation.	Logically deconstructs a problem and designs a well-considered and clear physics investigation.	Deconstructs a problem and designs a considered and generally clear physics investigation..	Prepares a basic deconstruction of a problem and an outline of a physics investigation.	Attempts a simple deconstruction of a problem and a procedure for a physics investigation.	!
9.—Results: Present the data collected in a table of results. A line graph with a line or curve of best fit OR a column graph. Max-min bars will show the precision of your measurements. Describe the trends and relationships as shown by your data. The graph of the sinusoidal function $f(x) = 1 + 2 * \sin(x)$ on the interval $0 \leq x \leq 2\pi$ is shown below.	!	Obtains, records, and represents data, using appropriate conventions and formats accurately and highly effectively.	Obtains, records, and represents data, using appropriate conventions and formats mostly accurately and effectively.	Obtains, records, and represents data, using generally appropriate conventions and formats with some errors but generally accurately and effectively.	Obtains, records, and represents data, using conventions and formats inconsistently, with occasional accuracy and effectiveness.	Attempts to record and represent some descriptive data, with limited accuracy or effectiveness.	!		

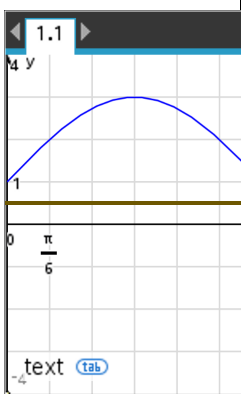


Limaçon Curve

Student Activity



The x scale is $\pi/6$.



Graph the limaçon given by $r_1(\theta) = 1 + 2\sin(\theta)$. Trace along the graph of the limaçon. Observe the change in the r and θ values. Explain why the polar curve $r = 1 + 2\sin(\theta)$ has an inner loop in the interval $\frac{7\pi}{6} \leq \theta \leq \frac{11\pi}{6}$ (θ between ≈ 3.7 rad and ≈ 5.7 rad).

Discussion: Analyse and evaluate precision of your data and a description of sources of errors.

IAE3

Systematically analyses and interprets data and

Logically analyses and interprets

Undertakes some analysis and interpretation

Describes data and undertakes some basic

Attempts to describe results

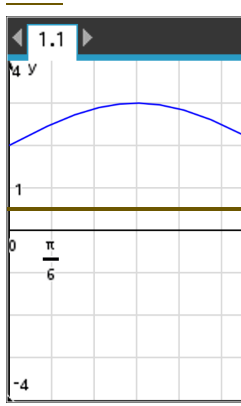
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Limaçon Curve

Student Activity



<u>Conclusion includes a clear and concise statement based on the data; supports or refutes the hypothesis. Any relationships established.</u>			<u>evidence to formulate logical conclusions with detailed justification.</u>	<u>data and evidence to formulate suitable conclusions with reasonable justification.</u>	<u>of data and evidence to formulate generally appropriate conclusions with some justification.</u>	<u>interpretation to formulate a basic conclusion.</u>	<u>and/or interpret data to formulate a basic conclusion.</u>	
<u>Discussion: Evaluate your procedure and describe and justify three improvements you could make to the experiment design.</u>		IAE4		<u>Critically and logically evaluates procedures and their effects on data.</u>	<u>Logically evaluates procedures and their effects on data.</u>	<u>Evaluates procedures and some of their effects on data.</u>	<u>Attempts to evaluate procedures or suggest an effect on data.</u>	<u>Acknowledges that procedures affect data.</u>
10. <u>Demonstration</u> The graph of physics understanding throughout the report, particularly $\sin$ and $\cos$ functions. $f(x) = 2 + 1 \cdot \sin(x)$ on the introduction and interval $0 \leq x \leq 2\pi$ is shown below. The x-scale is $\pi/6$. $2 + 2 \cdot \sin$ 		K A 1	<u>Demonstrates deep and broad knowledge and understanding of a range of physics concepts.</u>	<u>Demonstrates some depth and breadth of knowledge and understanding of a range of physics concepts.</u>	<u>Demonstrates knowledge and understanding of a general range of physics concepts.</u>	<u>Demonstrates some basic knowledge and partial understanding of physics concepts.</u>	<u>Demonstrates limited recognition and awareness of physics concepts.</u>	!



Limaçon Curve

Student Activity



Graph the interpretation of the limaçon given by $r = 2 + 1 \sin(\theta)$. Trace along the graph of results, the limaçon. Observe the change in the r and θ values. Explain why the polar curve does not contain the point located at the pole.									
<u>Spelling, grammar, sentence and paragraph structure.</u> <u>Appropriate use of physics terminology</u> <u>Method summarised</u> <u>Neat and clear presentation in an appropriate format.</u>			KA4		<u>Communicate s knowledge and understanding of physics coherently with highly effective use of appropriate terms, conventions, and representations.</u>	<u>Communicates knowledge and understanding of physics mostly coherently with effective use of appropriate terms, conventions, and representations.</u>	<u>Communicate s knowledge and understanding of physics generally effectively, using some appropriate terms, conventions, and representations.</u>	<u>Communicates basic physics information, using some appropriate terms, conventions, and/or representations.</u>	<u>Attempts to communicate information about physics.</u>